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A Trivially Short SAND Report (justified, report-style)

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ABSTRACT

So abstract it hurts.

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LIST OF TABLES

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1. INTRODUCTION

Very glad to meet you.

1.1. A Section

Underpromising, and hopefully over-delivering.

1.2. Another Section

Look at us go!

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2. BACKGROUND

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3. OBJECTIVE

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4. DATA

4.1. C Frame Modal

4.1.1. cframe_modal/jobs_coarse.iga

```
1 {
2   "coreform_iga_version": "2025.7",
3   "data_output_file": "cf_iga_data_output.json",
4   "data_output_interval": "intervals_1",
5   "flex_models": [
6     {
7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [
28    {
29      "constant": {
30        "value": 1
31      },
32      "label": "functions_1"
33    }
34  ],
35  "intervals": [
36    {
37      "label": "intervals_1",
38      "start_step": 0,
39      "step_increment": 1,
40      "stop_step": 20,
41      "time_increment": 1
42    }
43  ],
44  "label": "iga_1",
45  "materials": [
46    {
47      "elastic": {
48        "poissons_ratio": 0.29,
49        "youngs_modulus": 28000000
50      },
51      "label": "materials_1",
52      "mass_density": 0.000735
53    }
54  ],
55  "procedures": [
56    {
57      "label": "procedures_1",
58      "structural_modal_analysis": {
59        "boundary_conditions": [
60          "boundary_conditions_1"
61        ],
62        "flex_model": "flex_models_1",
63        "modal_solution_method": "modal_solution_methods_1",
64        "outputs": [
65          "outputs_1"
66        ]
67      }
68    }
69  ]
70 }
```

```

68 }
69 ],
70 "structural_modal_analysis_definitions": {
71   "boundary_conditions": [
72     {
73       "components": [
74         "x",
75         "y",
76         "z"
77       ],
78       "label": "boundary_conditions_1",
79       "penalty": {
80         "dimensionless": true,
81         "value": 10
82       },
83       "quadrature_options": {
84         "optimization_lawson_hanson_solve_completion": true,
85         "optimization_method": "tet_pruning",
86         "optimization_residual_tolerance": 1.0000000000000002e-12,
87         "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88         "quadrature": "QP1"
89       },
90       "set": "fixed"
91     }
92   ],
93   "modal_solution_methods": [
94     {
95       "frequency_extraction": {
96         "maximum_frequency": 4000,
97         "minimum_frequency": 1
98       },
99       "label": "modal_solution_methods_1"
100     }
101   ],
102   "outputs": [
103     {
104       "field": {
105         "database_name": "cframe_modal_coarse",
106         "external_surfaces_only": false,
107         "include_integration_point_output": false,
108         "interval": "intervals_1",
109         "part": "part_1",
110         "variables": {
111           "displacement": [
112             "all"
113           ]
114         }
115       },
116       "label": "outputs_1",
117       "tessellate": true
118     }
119   ]
120 }
121 }

```

4.1.2. cframe_modal/jobs_fine.iga

```

1 {
2   "coreform_iga_version": "2025.7",
3   "data_output_file": "cf_iga_data_output.json",
4   "data_output_interval": "intervals_1",
5   "flex_models": [
6     {
7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [

```

```

28   {
29     "constant": {
30       "value": 1
31     },
32     "label": "functions_1"
33   }
34 ],
35 "intervals": [
36   {
37     "label": "intervals_1",
38     "start_step": 0,
39     "step_increment": 1,
40     "stop_step": 20,
41     "time_increment": 1
42   }
43 ],
44 "label": "iga_1",
45 "materials": [
46   {
47     "elastic": {
48       "poissons_ratio": 0.29,
49       "youngs_modulus": 28000000
50     },
51     "label": "materials_1",
52     "mass_density": 0.000735
53   }
54 ],
55 "procedures": [
56   {
57     "label": "procedures_1",
58     "structural_modal_analysis": {
59       "boundary_conditions": [
60         "boundary_conditions_1"
61       ],
62       "flex_model": "flex_models_1",
63       "modal_solution_method": "modal_solution_methods_1",
64       "outputs": [
65         "outputs_1"
66       ]
67     }
68   }
69 ],
70 "structural_modal_analysis_definitions": {
71   "boundary_conditions": [
72     {
73       "components": [
74         "x",
75         "y",
76         "z"
77       ],
78       "label": "boundary_conditions_1",
79       "penalty": {
80         "dimensionless": true,
81         "value": 10
82       },
83       "quadrature_options": {
84         "optimization_lawson_hanson_solve_completion": true,
85         "optimization_method": "tet_pruning",
86         "optimization_residual_tolerance": 1.0000000000000002e-12,
87         "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88         "quadrature": "QP1"
89       },
90       "set": "fixed"
91     }
92   ],
93   "modal_solution_methods": [
94     {
95       "frequency_extraction": {
96         "maximum_frequency": 4000,
97         "minimum_frequency": 1
98       },
99       "label": "modal_solution_methods_1"
100     }
101   ],
102   "outputs": [
103     {
104       "field": {
105         "database_name": "cframe_modal_fine",
106         "external_surfaces_only": false,
107         "include_integration_point_output": false,
108         "interval": "intervals_1",
109         "part": "part_1",
110         "variables": {
111           "displacement": [
112             "all"
113           ]
114         }
115       },
116       "label": "outputs_1",
117       "tessellate": true
118     }
119   ]

```

```
120 }
121 }
```

4.1.3. cframe_modal/jobs_fine_x2.iga

```
1 {
2   "coreform_iga_version": "2025.7",
3   "data_output_file": "cf_iga_data_output.json",
4   "data_output_interval": "intervals_1",
5   "flex_models": [
6     {
7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [
28    {
29      "constant": {
30        "value": 1
31      },
32      "label": "functions_1"
33    }
34  ],
35  "intervals": [
36    {
37      "label": "intervals_1",
38      "start_step": 0,
39      "step_increment": 1,
40      "stop_step": 20,
41      "time_increment": 1
42    }
43  ],
44  "label": "iga_1",
45  "materials": [
46    {
47      "elastic": {
48        "poissons_ratio": 0.29,
49        "youngs_modulus": 28000000
50      },
51      "label": "materials_1",
52      "mass_density": 0.000735
53    }
54  ],
55  "procedures": [
56    {
57      "label": "procedures_1",
58      "structural_modal_analysis": {
59        "boundary_conditions": [
60          "boundary_conditions_1"
61        ],
62        "flex_model": "flex_models_1",
63        "modal_solution_method": "modal_solution_methods_1",
64        "outputs": [
65          "outputs_1"
66        ]
67      }
68    }
69  ],
70  "structural_modal_analysis_definitions": {
71    "boundary_conditions": [
72      {
73        "components": [
74          "x",
75          "y",
76          "z"
77        ],
78        "label": "boundary_conditions_1",
79        "penalty": {
```

```

80     "dimensionless": true,
81     "value": 10
82 },
83 "quadrature_options": {
84     "optimization_lawson_hanson_solve_completion": true,
85     "optimization_method": "tet_pruning",
86     "optimization_residual_tolerance": 1.0000000000000002e-12,
87     "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88     "quadrature": "QP1"
89 },
90 "set": "fixed"
91 }
92 ],
93 "modal_solution_methods": [
94     {
95         "frequency_extraction": {
96             "maximum_frequency": 4000,
97             "minimum_frequency": 1
98         },
99         "label": "modal_solution_methods_1"
100     }
101 ],
102 "outputs": [
103     {
104         "field": {
105             "database_name": "cframe_modal_fine_x2",
106             "external_surfaces_only": false,
107             "include_integration_point_output": false,
108             "interval": "intervals_1",
109             "part": "part_1",
110             "variables": {
111                 "displacement": [
112                     "all"
113                 ]
114             }
115         },
116         "label": "outputs_1",
117         "tessellate": true
118     }
119 ]
120 }
121 }

```

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5. METHODS

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6. RESULTS

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7. CONCLUSION

That is, as they say, all, folks.

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APPENDIX A. Appendices

Include any supplementary material, such as additional data, calculations, or detailed explanations of methods that are too lengthy for the main sections.



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